

■ TerraSpectra (Project 1): Complete Team Research, Preparation & Action Guide

This guide details everything your 4-member team needs to research, study, collect, and execute for Week 1 of Project 1: TerraSpectra (Hyperspectral Crop Disease Forecasting).

■ SECTION 1: Theoretical Research & Background Study (What to Learn)

All team members should read and understand these core concepts before writing code:

1.1 Remote Sensing & Hyperspectral Physics

- **Why RGB/NDVI Fails:** Standard 3-band RGB and 4-band NDVI imagery only detect plant stress after leaves turn yellow or brown (chlorosis/necrosis). At this point, plant cell structures are permanently destroyed.
- **The Hyperspectral Advantage:** Hyperspectral sensors capture **200+ contiguous narrow bands** (400 nm – 2500 nm).
- **Key Spectral Regions to Study:**
 - **Visible (400 nm – 700 nm):** Absorbed by Chlorophyll-a (675 nm) and Chlorophyll-b (650 nm).
 - **Red Edge (680 nm – 750 nm):** Steep incline zone. When fungal blight infects a crop, cell wall damage causes a **leftward shift** (toward shorter wavelengths) **3 weeks before visual symptoms manifest**.
 - **Near-Infrared / NIR (700 nm – 1100 nm):** Reflects off leaf spongy mesophyll. Infection causes reflection collapse.
 - **Short-Wave Infrared / SWIR (1100 nm – 2500 nm):** Detects plant moisture. Absorption troughs occur at 1450 nm and 1940 nm.

1.2 Data File Formats & GIS Coordinates

- **HDF5 (.h5 / .hdf5):** Stores multi-dimensional arrays (tensors) with metadata. Learn `h5py` in Python to inspect dataset keys, attributes, and slicing.
- **GeoTIFF (.tif):** Raster image tagged with geospatial parameters (GeoKeys). Maps pixel (x, y) to real-world Longitude/Latitude.
- **Coordinate Reference Systems (CRS):**
 - **EPSG:4326 (WGS 84):** Decimal degrees (Latitude, Longitude). Used for GPS & backend payloads.
 - **EPSG:3857 (Web Mercator):** Coordinates in meters. Used by web mapping engines (Mapbox, Google Maps).

■ SECTION 2: Mathematical Research & Formulas to Master

2.1 Principal Component Analysis (PCA) on 3D Cubes

Hyperspectral data has extreme redundancy between adjacent bands. PCA reduces 224 bands down to 32 principal components while retaining **>98.5% of variance**.

1. **Reshape 3D Data Cube:** $C \in \mathbb{R}^{H \times W \times B} \rightarrow X \in \mathbb{R}^{(H \cdot W) \times B}$
2. **Mean Centering & Standardization:** $\bar{X} = X - \mu$
3. **Covariance Matrix:** $\Sigma = \frac{1}{N-1} \bar{X}^T \bar{X} \in \mathbb{R}^{B \times B}$
4. **Eigenvalue Decomposition:** $\Sigma V = V \Lambda \quad (\text{Eigenvalues } \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_B)$
5. **Cumulative Explained Variance Ratio:** $\text{EVR}(k) = \frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^B \lambda_j} \geq 0.985$
6. **Projection Back to 3D Tensor:** $Z = \bar{X} V_k \in \mathbb{R}^{(H \cdot W) \times k} \rightarrow C_{\text{reduced}} \in \mathbb{R}^{H \times W \times k}$

SECTION 3: Data Assets, Accounts & API Keys to Collect

Before building, the team must acquire the following items:

1. **Mapbox Access Token (Free Developer Account):**
2. Register at mapbox.com.
3. Copy the default public access token (pk.eyJ1. . .).

Add to frontend/.env file as VITE_MAPBOX_ACCESS_TOKEN.

Satellite Datasets & Sample Files (To Research & Download):

6. **NASA AVIRIS-NG Airborne Data:** Download sample .hdr / binary spectral cubes from aviris.jpl.nasa.gov.
7. **USGS EarthExplorer EO-1 Hyperion:** Access spaceborne 222-band satellite scenes via earthexplorer.usgs.gov.
8. **Synthetic Data Generator:** Run the local generator script `scripts/generate_mock_data.py` to create a 224-band HDF5 data cube (`data/mock_farm_datacube.h5`).

SECTION 4: Software Stack & Environment Setup

1. Python Environment (Backend & Data Science)

Ensure Python 3.10+ or 3.11+ is installed. Install required packages:

```
pip install rasterio h5py scikit-learn fastapi uvicorn torch torchvision
```

2. Node.js Environment (Frontend GIS Interface)

Ensure Node.js 18+ is installed. Install frontend dependencies:

```
cd frontend
npm install react react-dom @deck.gl/core @deck.gl/layers @deck.gl/react mapbox-gl react-map-gl lucide-react
```

■ SECTION 5: Week-1 Task Allocation Matrix (For the 4 Team Members)

■■■ Member 1: Systems & API Architecture Lead

- **What to Research:** FastAPI async route handlers, Pydantic v2 schemas, CORS middleware, REST endpoint design.
- **Week 1 Deliverables:**
 - Set up backend/main.py with FastAPI server.
 - Implement /health and /api/v1/pca/reduce REST API routes.
 - Test API using Swagger docs (<http://localhost:8000/docs>).

■■■ Member 2: Geospatial & Data Engineer

- **What to Research:** Rasterio datasets, h5py file indexing, PCA dimensionality reduction using sklearn.decomposition.PCA.
- **Week 1 Deliverables:**
 - Run scripts/generate_mock_data.py to generate mock_farm_datacube.h5.
 - Build backend/app/services/data_pipeline.py to ingest HDF5 & GeoTIFF files.
 - Build backend/app/services/pca_reducer.py to perform 224 $\rightarrow$ 32 component PCA reduction.

■■■ Member 3: Deep Learning & PyTorch Specialist

- **What to Research:** PyTorch 3D Convolutional layers (nn.conv3d), input tensor dimension requirements (Batch, Channels, Depth, Height, Width).
- **Week 1 Deliverables:**
 - Study 3D-CNN kernel mechanics for spatial-spectral feature extraction.
 - Draft baseline PyTorch model class Hybrid3DCNNViT.
 - Measure PyTorch memory requirements for 32-component PCA tensors.

■■■ Member 4: Frontend GIS & React UI Developer

- **What to Research:** React 18, Vite, Deck.gl v8+ layers (GeoJsonLayer, BitmapLayer), Mapbox GL JS satellite map styles.
- **Week 1 Deliverables:**
 - Obtain Mapbox public access token.
 - Set up React + Vite + TypeScript frontend in frontend/.

- Build `MapContainer.tsx` rendering Mapbox satellite basemap + Deck.gl Iowa Farm Zone Alpha boundary layer.

■ SECTION 6: How to Test & Verify Everything

Test Synthetic Data Generation: `bash python3 scripts/generate_mock_data.py` *Verify:* Creates `data/mock_farm_datacube.h5` (~198 MB) and `data/mock_farm_preview.tif`.

Test Data Pipeline & PCA Reduction: `bash python3 -c " import sys; sys.path.append('backend')
from app.services.data_pipeline import HyperspectralDataLoader from app.services.pca_reducer
import SpectralPCAReducer cube, w, meta =
HyperspectralDataLoader.load_hdf5_cube('data/mock_farm_datacube.h5') reducer =
SpectralPCAReducer(n_components=32) red_cube, stats = reducer.fit_transform_cube(cube)
print('Success! Shape:', red_cube.shape) "`

Test Frontend Development Server: `bash cd frontend npm run dev` *Verify:* Open `http://localhost:5173` to see the interactive Mapbox satellite map with the Iowa Farm Zone Alpha layer.